MIL-S-6785D
1 November 1968
Superseding
MIL-S-6785C
11 October 1955

MILITARY SPECIFICATION
SOLENOID, REMOTE GUN FIRING, TYPE G-9B

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for one type of remote gun firing solenoid, designated as Type G-9B.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery.

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U. S. Military Property.

DRAWINGS

U. S. Air Force

5404628	Solenoid, Remote Gun Firing, Type G-9B

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5945

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3. REQUIREMENTS

3.1 Preproduction samples. This specification requires the submission of preproduction samples for preproduction testing. See Section 4 for details.

3.2 Materials. Materials shall be as specified herein. When materials are used which are not specifically designated, they shall be entirely suitable for the purpose.

3.3 Design. The requirements of Drawing 5404628 are applicable as requirements of this specification.

3.3.1 Voltage. The normal operating voltage shall be 28v dc.

3.3.2 Terminal guard. A cap shall be provided for the connector to protect the threaded end against damage when the retainer is not attached.

3.4 Performance. The solenoid shall meet the following performance tests:

(a) Current consumption. (See 4.4.6.2)

(b) Operation. (See 4.4.6.3)

(c) Time of operation. (See 4.4.5.1)

(d) Temperature rise. (See 4.4.5.2)

(e) Immersion. (See 4.4.5.3)

(f) Impact. (See 4.4.5.4)

(g) Dimensional. (See 4.4.5.5)

3.5 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.6 Workmanship. Workmanship shall be consistent with the requirements specified herein and of the quality necessary to produce solenoid assemblies free from all defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed

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necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order for a period of two years.

4.2 Classification of tests. The inspection and testing of the solenoids shall be as follows:

a. Qualification tests (See 4.3)

b. Acceptance tests (See 4.4)

(1) Sampling tests (See 4.4.5)

(2) Individual tests (See 4.4.6)

(a) Examination of product (See 4.4.6.1)

(b) Current consumption (See 4.4.6.2)

(c) Operation (See 4.4.6.3)

c. Preproduction tests (See 4.3)

4.2.1 Qualification test samples. Qualification test samples shall consist of solenoids selected at random and representative of production equipment. The samples shall be identified with the manufacturer's part number.

4.2.2 Tests. Qualification tests of solenoids shall consist of all the tests in 4.3. The units shall be tested at a laboratory approved by the procuring activity, or when so stated in the contract, at the manufacturer's plant under the supervision of the procuring activity. A certified test report of qualification tests in accordance with MIL-STD-831 signed by a responsible company official shall be prepared and three copies forwarded to the qualifying agency. The qualification test report shall state whether or not the qualification samples were production units.

4.3 Preproduction tests.

4.3.1 Preproduction test sample. One or more solenoids as specified in the contract shall be tested for design approval by the procuring activity or, when so specified in the contract, at the contractor's plant under the supervision of the procuring activity.

4.3.2 Gun firing tests. A 10,000 round gun firing test shall be conducted without any resultant damage to the solenoid.

4.4 Acceptance tests.

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4.4.1 Acceptance test samples. One solenoid shall be selected at random from each lot of 1000, or fraction thereof, on the order and subjected to the sampling tests. These tests shall be in addition to the individual tests specified herein.

4.4.2 Gun fire test samples. One solenoid from each lot of 3000, or fraction thereof, shall be subjected to a 2500 round gun firing test on a free firing caliber .50 M3 machine gun.

4.4.3 Rejection and retest. When one or more items from a lot fail to meet the specification, acceptance of all items in the lot shall be withheld until the extent and cause of failure has been determined. The contractor shall explain fully to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the sampling tests shall be repeated.

4.4.4 Test conditions. Unless otherwise specified, the solenoids shall be subjected to the tests under the conditions specified herein.

4.4.5 Sampling tests.

4.4.5.1 Time of operation. At a starting and ambient temperature of 20°C (68°F) and with an applied electromotive force of not more than 28v dc, the sample solenoid shall be tested to determine that the time required for the solenoid plunger to travel from its fully open to its fully closed position shall not exceed 0.04 second, and also that the time required for the solenoid plunger to travel from its fully closed position (air gap closed) to its fully open position, after breaking the circuit, shall not exceed 0.04 second.

4.4.5.2 Temperature rise. The solenoid shall be subjected to a constant electromotive force of 28v dc for 5 minutes in still air at a starting and ambient temperature of 20°C (68°F). There shall be no damage to the solenoid as a result of this test. After cooling to the above temperature, the solenoid shall satisfactorily operate as specified herein.

4.4.5.3 Immersion test. The solenoid shall be immersed in fresh water for a period of 24 hours. It shall then be dried and tested to determine compliance with operation and time of operation tests specified herein.

4.4.5.4 Impact test. An impact test of approximately 0.75 foot pounds shall be applied to the solenoid adjusting cap by dropping a weight of 0.75 pounds from a height of 1 foot on the adjusting cap. Plunger protrusion upon this impact shall not exceed 0.035 inch. The impact shall be repeated at least 4 times, with the center of impact at different parts of the cap each time for verification.

4.4.5.5 Dimensional test. The dimensional test shall be conducted in accordance with the requirements for dimension "C" as specified in the notes on drawing 5404628.

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4.4.6 Individual tests. Each solenoid shall be subjected to the following tests:

4.4.6.1 Examination of product. While performing the tests required by 4.4.6.2 and 4.4.6.3, each solenoid shall be inspected with regard to material, workmanship, markings, and dimensions to determine its conformity with the requirements of 3.2, 3.5, 3.6, and drawing 5404628.

4.4.6.2 Current consumption. Each solenoid shall be tested to determine that the current does not exceed 6.2 amperes at 28v dc.

4.4.6.3 Operation. At a starting and ambient temperature of 20°C (68°F), and with a 0.050 inch air gap, the solenoid shall develop a minimum static effort at 70 pounds with an electromotive force of 28v applied to the terminal.

4.5 Inspection of preservation, packaging, packing, and marking for shipment and storage. Preservation, packaging, packing, and marking for shipment and storage shall be inspected to determine conformance with the documents referenced in Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Unless otherwise specified, preservation, packaging, and packing shall be in accordance with MIL-E-17555.

5.2 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The type G-9B solenoid, as covered by this specification, is intended for use in airacraft for the electric control of the caliber .50 M3 machine gun.

6.2 Ordering data. Procurement documents should specify the following:

a. Name, type, and model designation of the equipment.

b. Title, number, and date of this specification.

c. List of components being procured, by name and type designation of each component, as given on the current Parts Control List.

d. Selection of applicable levels of preservation, packaging, and packing required.

e. Number of preproduction samples and location of tests.

6.3 Gun fire tests. Gun fire tests as specified in 4.3.2 and 4.4.2 shall be conducted by the procuring activity, due to the lack of firing range facilities at most manufacturing plants.

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Custodians:

ARMY - MU

NAVY - AS

AIR FORCE - 85

Review activities:

Army - MU
Navy - AS, MC
Air Force - 84
Air Force - 85

Preparing Activity:

Air Force - 85

Project Number:

5945-0147

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4.4 First article. First article solenoids shall be furnished for the inspections specified in 4.8.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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